

# Combinations of Scheimpflug tomography, ocular coherence tomography and air-puff tonometry improve the detection of keratoconus

Nan-Ji Lu<sup>a,b,c,d</sup>, Carina Koppen<sup>c,e</sup>, Farhad Hafezi<sup>a,b,d,f,g,h</sup>, Sorcha Ní Dhubhghaill<sup>c,e</sup>, Ioannis M. Aslanides<sup>a,b,i</sup>, Qin-Mei Wang<sup>a,b,\*</sup>, Le-Le Cui<sup>a,b,\*</sup>, Jos J. Rozema<sup>c,e</sup>

<sup>a</sup> National Clinical Research Center for Ocular Diseases, Eye Hospital, Wenzhou Medical University, Wenzhou, 325027, China

<sup>b</sup> State Key Laboratory of Ophthalmology, Optometry and Visual Science, Eye Hospital, Wenzhou Medical University, Wenzhou, 325027, China

<sup>c</sup> Faculty of Medicine and Health Sciences, University of Antwerp, Wilrijk, Belgium

<sup>d</sup> ELZA Institute, Dietikon, Switzerland

<sup>e</sup> Department of Ophthalmology, Antwerp University Hospital, Edegem, Belgium

<sup>f</sup> Laboratory of Ocular Cell Biology, Center for Applied Biotechnology and Molecular Medicine, University of Zurich, Zurich, Switzerland

<sup>g</sup> Faculty of Medicine, University of Geneva, Geneva, Switzerland

<sup>h</sup> USC Roski Eye Institute, University of Southern California, Los Angeles, CA, USA

<sup>i</sup> Emmetropia Mediterranean Eye Institute, Heraklion, Crete, Greece

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## ABSTRACT

**Purpose:** To determine whether combinations of devices with different measuring principles, supported by artificial intelligence (AI), can improve the diagnosis of keratoconus (KC).

**Methods:** Scheimpflug tomography, spectral-domain optical coherence tomography (SD-OCT), and air-puff tonometry were performed in all eyes. The most relevant machine-derived parameters to diagnose KC were determined using feature selection. The normal and forme fruste KC (FFKC) eyes were divided into training and validation datasets. The selected features from a single device or different combinations of devices were used to develop models based on random forest (RF) or neural networks (NN) trained to distinguish FFKC from normal eyes. The accuracy was determined using receiver operating characteristic (ROC) curves, area under the curve (AUC), sensitivity, and specificity.

**Results:** 271 normal eyes, 84 FFKC eyes, 85 early KC eyes, and 159 advanced KC eyes were included. A total of 14 models were built. Air-puff tonometry had the highest AUC for detecting FFKC using a single device (AUC = 0.801). Among all two-device combinations, the highest AUC was accomplished using RF applied to selected features from SD-OCT and air-puff tonometry (AUC = 0.902), followed by the three-device combination with RF (AUC = 0.871) with the best accuracy.

**Conclusion:** Existing parameters can precisely diagnose early and advanced KC, but their diagnostic ability for FFKC could be optimized. Applying an AI algorithm to a combination of air-puff tonometry with Scheimpflug tomography or SD-OCT could improve FFKC diagnostic ability. The improvement in diagnostic ability by combining three devices is modest.

## 1. Introduction

Keratoconus (KC) is a bilateral, asymmetrical disorder characterized by the steepening and thinning of the cornea, leading to an irregular astigmatism that impairs vision [1]. Early diagnosis can help surgeons monitor the disease progression and perform corneal cross-linking (CXL) early to halt progression [2]. It also assists screening laser vision

correction (LVC) candidates for ectasia risks.

Pathophysiologically, KC is thought to start as a regional impairment of the corneal biomechanics that leads to a remodelling of the shape [3]. Over the past decades, diagnostic tools have become more diverse, from Placido ring topography, which precisely measures the anterior cornea curvature, to Scheimpflug tomography, which also provides posterior surface information. Other high resolution technologies, such as very

\* Corresponding authors at: Eye Hospital of Wenzhou Medical University, National Clinical Research Center for Ocular Diseases, 270 West Xueyuan Road, Wenzhou, Zhejiang, China.

E-mail addresses: [wqm97028@wmu.edu.cn](mailto:wqm97028@wmu.edu.cn) (Q.-M. Wang), [cll@eye.ac.cn](mailto:cll@eye.ac.cn) (L.-L. Cui).

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high frequency (VHF) digital ultrasound and optical coherence tomography (OCT), enabled measurements of the corneal epithelial thickness [4], another parameter that may assist KC diagnosis [5,6]. To capture the corneal biomechanical properties, the calibrated non-contact tonometer couples with an ultra-high-speed Scheimpflug camera and thus assesses the dynamic changes in corneal deformation [7–9]. Most devices provide parameters for KC based on regression [10–13] or artificial intelligence [8,14–17], while others have parameters that combine data from two different measuring principles [18–21], each with claims of high diagnostic capabilities.

Although tomographic and biomechanical devices are capable of capturing different properties of the cornea, it is impractical and uneconomical to use every available device to make a diagnosis. It is therefore important to determine the value of the added information needed for diagnosis, whether provided by a single device or a combination of three devices with different working principles. To this end, this study compares the diagnostic performance of a number of combinations of clinical devices and their roles in the diagnosis for various degrees of KC.

## 2. Methods

This retrospective study was approved by the Ethics Committee of the Hospital and adhered to the tenets of the Declaration of Helsinki.

### 2.1. Patients groups and inclusion criteria

This study considered four patient groups: normal controls, forme fruste keratoconus (FFKC), early-stage KC (EKC), and advanced KC (AKC) as defined by the criteria below [21].

Normal controls were recruited from patients seeking LVC surgery in the Hospital, excluding those with ocular or systemic abnormalities, prior ocular surgery. Inclusion criteria were a stable corrected distance visual acuity (CDVA)  $\geq 20/20$  for 2 years with normal Scheimpflug tomography (Pentacam HR, Oculus Optikgeräte, Wetzlar, Germany, software version 1.25r12) confirmed by three doctors before LVC surgery. To exclude those developing post-LVC ectasia, patients were followed for at least 3 years postoperatively. Only one randomly selected eye was included per person in the control group.

The diagnosis of KC was clinical and confirmed by three doctors, which ordinarily required at least 1 typical slit lamp finding (Fleischer ring, Vogt striae, or central thinning) and/or at least 1 typical sign of KC on Scheimpflug tomography, such as decreased thinnest pachymetry, abnormal pachymetry distribution, skewed asymmetric bowtie/inferior steep [SAB/IS], or increased inferior steepness. For the three KC groups, only the eye with mildest impairment was included in each KC patient.

The criteria for the FFKC group were: 1) a contralateral eye diagnosed with KC, 2) best CDVA  $\geq 20/20$ , 3) no KC signs at slit-lamp, 4)  $K_{\max} < 47.4$  diopters (D), 5) thinnest pachymetry  $\geq 480$   $\mu\text{m}$  obtained in Pentacam HR, “normal” tomography with the difference between the  $K_{\max}$  values in the inferior and superior areas at 3 mm (I-S Value)  $< 1.4$  D, and no SAB/IS [21,22]. For the EKC group, the criteria were: 1)  $K_{\max} < 48.5$  D and smallest thickness greater than 480  $\mu\text{m}$ , 2) best CDVA  $\geq 16/20$ , and 3) no central scars and fewer than two slit lamp findings. Finally, the criteria for the AKC group were KC eyes with parameters exceeding those of the FFKC and EKC groups (i.e.,  $K_{\max} \geq 48.5$  D, smallest thickness  $< 480$   $\mu\text{m}$ , best CDVA  $< 16/20$ ).

### 2.2. Data acquisition and evaluated parameters

Patients were asked to discontinue soft contact lenses for at least 2 weeks before the examination, or at least 4 weeks in cases of rigid gas-permeable contact lenses wear. Patients who could not comply were eliminated from the analysis.

### 2.3. Scheimpflug tomography

Tomography measurements were performed with a Pentacam HR. The machine-derived parameters were exported, including the maximum/ average/ minimum pachymetry progression ( $RPI_{\text{Max/Mid/Min}}$ ), averaged pachymetry ( $D_{0\text{mm}} - D_{10\text{mm}}$  Pachy), index of surface variance (ISV), index of vertical asymmetry (IVA), keratoconus index (KI), central keratoconus index (CKI), index of height asymmetry (IHA), and index of height decentration (IHD).

### 2.4. Spectral-Domain OCT

The measurement was performed with the RTVue-XR Spectral-Domain OCT (Optovue, Inc., Fremont, CA, USA) [23]. The output consists of thickness maps for the whole cornea (CT), corneal epithelium (ET), and stroma (ST) in the central (2 mm diameter), paracentral (2 to 5 mm), midperipheral (5 to 7 mm), and peripheral regions (7 to 9 mm). In the latter three regions, thickness is monitored in 8 equally spaced points along the median circumference of the region, including the temporal (T), superior-temporal (ST), superior (S), superior-nasal (SN), nasal (N), inferior-nasal (IN), inferior (I) and inferior-temporal (IT) positions.

### 2.5. Air-puff tonometry

Corneal biomechanics were measured by Corvis ST (Oculus Optikgeräte, Wetzlar, Germany, software version 1.6r2503), which recorded two categories' parameters: 1) independent parameters, including intraocular pressure (IOP), biomechanically corrected IOP (bIOP), pachymetry, and 35 dynamic corneal response (DCR) parameters; 2) Comprehensive parameters, including the stress-strain index (SSI), the Corvis biomechanical index (CBI), the Corvis biomechanical factor (CBI<sub>F</sub>), and the tomographic and biomechanical index (TBI) developed from DCR and topography parameters.

### 2.6. Artificial intelligence (AI)

The R software (version 4.2.0, R Foundation for Statistical Computing, Vienna, Austria) was used to develop AI models based on Random Forests (RF) and Neural Networks (NN). Before the training of AI models, feature selection was applied to the parameters exported from Pentacam HR, RTVue-XR, and Corvis ST using the *Boruta* package (Version 7.0.0) in R [24]. Two categories of parameters were excluded before feature selection: 1) parameters used in the groups' classification:  $K_{\max}$ , thinnest pachymetry, and I-S Value; 2) existing machine-derived parameters: the Pentacam random forest index (PRFI), Belin-Ambrósio deviation index (BADD), CBI, CBI beta, CBI<sub>F</sub>, and TBI.

### 2.7. Models

After feature selection, two algorithms to distinguish the FFKC group from the normal group were developed based on RF using the *randomForest* package (Version 4.6-14) and NN using the *neuralnet* package (Version 1.44.2) [25,26]. For the RF model, 600 decision trees were grown and combined to converge the out-of-bag error and improve prediction performance. For the NN model, an artificial NN was built on multi-layers of interconnected nodes, including 4 hidden layers and 5 hidden neurons, using a supervised learning algorithm. In each model, specific sub-models were developed based on the selected features representing combinations of the different devices: 1) only including features from Pentacam, RTVue-XR, or Corvis ST, respectively; 2) including features from device pairs (Pentacam with RTVue-XR, or Pentacam with Corvis ST, RTVue-XR with Corvis ST); 3) all three devices.

2.8. Validation

The dataset of normal and FFKC groups was randomly divided into a training and a validation set to determine the validity of the model. The training data constituted 60% of the total dataset and was used to train the models, whilst the remaining 40% of the total dataset formed the validation dataset used to evaluate the models' accuracy. The average value of the classification accuracy obtained after executing a 10-fold cross-validation in the validation dataset was recorded.

2.9. Statistical analysis

The statistical analysis was performed in R. The normality of the data was verified using the Shapiro-Wilks test. Descriptive statistics were presented as mean ± standard deviation. Analysis of variance (ANOVA) and Kruskal-Wallis H tests were used to analyze the differences between the four groups, followed by post-hoc tests with a Bonferroni correction. The 95% confidence intervals (CI) were calculated by the binomial exact. A value of  $P < 0.05$  was considered statistically significant. To determine the optimal cut-off values, sensitivity, and specificity, receiver operating characteristic (ROC) curves and area under the curve (AUC) as accuracy measures were used.

3. Results

3.1. Demographics

Demographic information is shown in Table 1. The parameter distributions and comparisons between patient groups are shown in Fig. 1.

Table 1  
Basic Demographic Information.

|                  | Unit    | Normal         | FFKC           | EKC            | AKC            |
|------------------|---------|----------------|----------------|----------------|----------------|
| Number           | (OD/OS) | 135/136        | 44/40          | 42/43          | 74/85          |
| Gender           | (M/F)   | 141/130        | 44/40          | 45/40          | 83/76          |
| Age              | Years   | 22.24 ± 6.05   | 23.58 ± 5.71   | 22.53 ± 7.22   | 21.79 ± 6.30   |
| K <sub>max</sub> | Dioptr  | 44.33 ± 1.50   | 44.30 ± 1.41   | 46.15 ± 2.81   | 56.05 ± 8.11   |
| Thinnest         | μm      | 541.45 ± 29.19 | 526.46 ± 26.77 | 507.39 ± 29.36 | 463.06 ± 39.41 |
| Pachymetry       |         |                |                |                |                |
| I-S Value        | Dioptr  | 0.19 ± 0.60    | 0.36 ± 0.51    | 1.53 ± 1.40    | 4.77 ± 3.22    |
| biOP             | mmHg    | 15.83 ± 2.19   | 14.83 ± 2.09   | 15.03 ± 2.45   | 13.84 ± 2.46   |
| DA Ratio Max     | mm      | 4.42 ± 0.42    | 4.62 ± 0.41    | 4.76 ± 0.57    | 5.85 ± 1.11    |
| 2 mm             |         |                |                |                |                |
| SPA1             | mmHg/mm | 112.91 ± 17.17 | 99.18 ± 17.04  | 93.93 ± 17.99  | 67.91 ± 23.12  |
| PRFI             |         | 0.12 ± 0.10    | 0.21 ± 0.17    | 0.66 ± 0.25    | 0.93 ± 0.18    |
| BADD             |         | 1.12 ± 0.59    | 1.41 ± 0.63    | 2.90 ± 1.59    | 8.16 ± 4.39    |
| SSI              |         | 0.87 ± 0.12    | 0.86 ± 0.15    | 0.85 ± 0.16    | 0.76 ± 0.17    |
| CBI              |         | 0.25 ± 0.27    | 0.39 ± 0.31    | 0.59 ± 0.34    | 0.90 ± 0.26    |
| CBiF             |         | 6.29 ± 0.38    | 6.03 ± 0.37    | 5.71 ± 0.48    | 4.68 ± 0.80    |
| TBI              |         | 0.24 ± 0.18    | 0.38 ± 0.25    | 0.87 ± 0.27    | 0.97 ± 0.14    |

FFKC = forme fruste keratoconus; EKC = early keratoconus; AKC = advanced keratoconus; I-S Value = the difference between average inferior and superior corneal powers 3 mm from the center of the cornea; biOP = biomechanically corrected intraocular pressure; DA Ratio Max 2 mm = deformation amplitude measured at 2 mm from the center; SPA1 = stiffness parameter at first applanation; PRFI, the Pentacam random forest index; BADD = the Belin-Ambrósio deviation index; SSI = the stress-strain index; CBI = the Corvis biomechanical index; CBiF = the Corvis biomechanical factor; TBI = the tomographic and biomechanical index.

There were no significant differences in right eye/left eye ratios, sex distribution, or age among all groups ( $P = 0.922, 0.999, \text{ and } 0.194$ , respectively).

3.2. The existing parameters ROC analysis

The comparisons of the normal group with the three KC groups, based on the parameters' AUC, TBI and the 3 highest-ranked parameters respectively obtained from Pentacam HR, RTVue-XR, and Corvis ST are listed in Table 2. Compared with the normal group vs. the AKC/EKC group, all parameters were far less performant in the normal group vs. the FFKC group.

3.3. Feature selection

Feature selection reduced the machine-derived parameters for Pentacam HR, RTVue-XR, and Corvis ST. These selected parameters are listed in Supplemental Materials I.

3.4. Artificial intelligence models and performance

A total of 14 models were developed, including 7 models built by RF and 7 using NN. The models' features, accuracy, sensitivity, specificity, and AUC are presented in Table 3. The model with the highest AUC was Model 6 (AUC = 0.902), based on RF applied to RTVue-XR and Corvis ST, followed by Model 7 (RF applied to all three devices; AUC = 0.871) and Model 14 (NN applied to all three devices; AUC = 0.869). The most accurate models were Models 14 (85.07%), 7 (84.33%), and 6 (84.29%). Comparing the AUC and accuracy of single-device models (Models 1–3 and 8–10), Models 3 and 10 from the Corvis ST performed best, followed by the RTVue-XR and Pentacam HR. The AUC and accuracy of the two-devices models was best in models including features from the Corvis ST (Models 5, 6, 12 and 13).

3.5. Comparisons of RF and NN

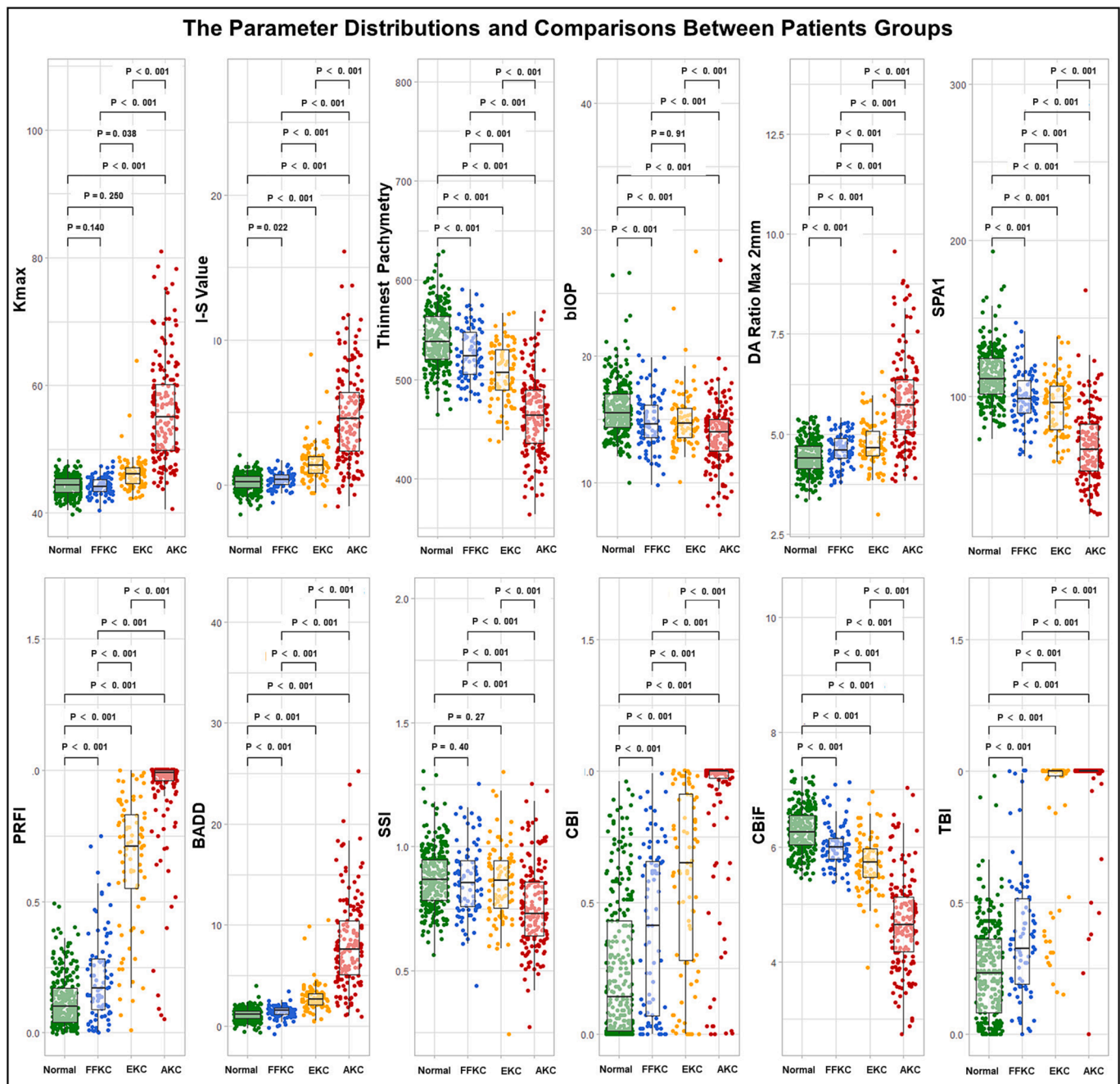
Comparing RF and NN applied to the same features, RF outperformed NN in all but two models: Model 11 had a higher AUC than Model 4 (0.681 vs. 0.658) and Model 14 had a higher accuracy and sensitivity than Model 7.

3.6. Comparisons between models and existing parameters

The ROC curves of the new models created by RF and NN and the 4 existing comprehensive parameters (PRFI, BADD, CBI, and TBI) are shown in Fig. 2a and Fig. 2b, respectively. Models 1 and 8 (selected features based on Pentacam HR only) performed similarly as PRFI and BADD; In contrast, Models 3 and 10 (selected features based on Corvis ST only), and Models 5 and 12 (selected features based on Pentacam HR and Corvis ST) outperformed CBI and TBI, respectively.

4. Discussion

A review of the literature revealed eight previous studies that combined different measuring principles to diagnose KC (Supplemental Material II) [18–21,27–30]. Their combination approaches can be categorized as: 1) Placido topography plus aberrometer [28]; 2) Scheimpflug tomography plus VHF digital ultrasound [20]; 3) Placido topography/ Scheimpflug tomography plus OCT [18,29,30]; 4) Scheimpflug tomography plus air-puff tonometry [19,27]; and 5) SD-OCT plus air-puff tonometry, as recently proposed by our research group using a different database [21]. The current study only considers the latter three for a number of reasons. Firstly, Scheimpflug tomography provides more information than Placido topography. Secondly, although VHF digital ultrasound may have high accuracy, it is slow and rarely used. Finally, aberrometers were eliminated as institutions



**Fig. 1.** The distributions and comparisons of patients' demographic characteristics among all groups. I-S Value = the difference between average inferior and superior corneal powers 3 mm from the center of the cornea; biOP = biomechanically corrected intraocular pressure; DA ratio 2 mm = the ratio between the central deformation and the average of peripheral deformation at 2.0 mm from center; SPA1 = stiffness parameter at first applanation; PRFI = the Pentacam random forest index; BADD = the Belin-Ambrosio deviation index; SSI = the stress-strain index; CBI = the Corvis biomechanical index; CBIF = the Corvis biomechanical factor; TBI = the tomographic and biomechanical index.

without a refractive unit are less likely to have one, making it less useful.

Here, three combinations of two devices with different measuring principles augmented by AI were reported. Combining Scheimpflug with SD-OCT (Models 4 and 11) did not improve the diagnostic ability for FFKC compared to Scheimpflug/ SD-OCT alone (Models 1, 2, 8, and 9). This might be because KC starts as a local weakening of the cornea, which does not show changes that can be detected by Scheimpflug/ SD-OCT. In contrast, when Hwang *et al.*, combined Scheimpflug tomography and SD-OCT using logistic regression, they obtained a highly performant model to distinguish asymmetric KC from normal eyes, which was superior to Scheimpflug or SD-OCT alone [29]. This contradictory

result could be due to the limited number of KC patients included in the study and the most of included highly asymmetric KC eyes were in the early KC stage but not in the forme fruste KC stage.

Among the two-device combinations, the FFKC diagnostic ability of Scheimpflug with air-puff tonometry (Models 5 and 12) and SD-OCT with air-puff tonometry (Models 6 and 13) was greatly improved compared to the single-device analyses. This again confirms the importance of biomechanical measurement in FFKC diagnosis, where corneal biomechanics abnormalities may occur in tomographically normal corneas [31]. The combination of air-puff tonometry with SD-OCT had a slightly better diagnostic ability for FFKC than the



**Table 2**

The Top 3 AUC Parameters from Each Device in the Normal Group vs. KC Groups.

| Device                                     | Variable                            | AUC   | 95% CI      | Cut-off Value | Sensitivity | Specificity |
|--------------------------------------------|-------------------------------------|-------|-------------|---------------|-------------|-------------|
| <b>the Normal Group vs. the AKC Group</b>  |                                     |       |             |               |             |             |
| Scheimpflug                                | TBI                                 | 0.987 | 0.973–1.000 | 0.912         | 0.996       | 0.956       |
| + Air-puff tonometry                       |                                     |       |             |               |             |             |
| Scheimpflug tomography                     | BAD-D                               | 0.986 | 0.974–0.999 | 2.355         | 0.993       | 0.943       |
| Scheimpflug tomography                     | ISV                                 | 0.986 | 0.975–0.997 | 29.500        | 0.989       | 0.937       |
| Scheimpflug tomography                     | PRFI                                | 0.985 | 0.971–0.999 | 0.505         | 1.000       | 0.956       |
| SD-OCT                                     | EPI Overall StdDev                  | 0.912 | 0.884–0.941 | 2.670         | 0.819       | 0.868       |
| SD-OCT                                     | EPI 5 mm SN-IT                      | 0.891 | 0.853–0.929 | 1.960         | 0.948       | 0.755       |
| SD-OCT                                     | Pachymetry Overall StdDev           | 0.886 | 0.851–0.921 | 22.07         | 0.830       | 0.818       |
| Air-puff tonometry                         | CBI beta                            | 0.952 | 0.926–0.979 | 2.044         | 0.993       | 0.881       |
| Air-puff tonometry                         | CBI                                 | 0.939 | 0.909–0.969 | 0.865         | 0.982       | 0.855       |
| Air-puff tonometry                         | Max_InverseRadius_mm <sup>-1</sup>  | 0.915 | 0.885–0.945 | 0.204         | 0.919       | 0.786       |
| <b>the Normal Group vs. the EKC Group</b>  |                                     |       |             |               |             |             |
| Scheimpflug                                | TBI                                 | 0.939 | 0.907–0.972 | 0.750         | 0.989       | 0.812       |
| + Air-puff tonometry                       |                                     |       |             |               |             |             |
| Scheimpflug tomography                     | PRFI                                | 0.957 | 0.927–0.988 | 0.365         | 0.974       | 0.847       |
| Scheimpflug tomography                     | BAD-D                               | 0.927 | 0.888–0.967 | 1.875         | 0.919       | 0.847       |
| Scheimpflug tomography                     | RPIMax                              | 0.887 | 0.842–0.933 | 1.555         | 0.911       | 0.765       |
| SD-OCT                                     | EPI 5 mm SN-IT                      | 0.765 | 0.703–0.827 | 1.325         | 0.904       | 0.553       |
| SD-OCT                                     | EPI 5 mm Superior-Inferior          | 0.718 | 0.651–0.786 | −0.605        | 0.683       | 0.647       |
| SD-OCT                                     | EPI 5 mm S-I                        | 0.694 | 0.621–0.766 | −0.620        | 0.734       | 0.588       |
| Air-puff tonometry                         | CBI beta                            | 0.823 | 0.767–0.879 | 0.343         | 0.823       | 0.718       |
| Air-puff tonometry                         | CBI                                 | 0.774 | 0.712–0.837 | 0.463         | 0.771       | 0.682       |
| Air-puff tonometry                         | DA_Ratio_Max_2mm                    | 0.686 | 0.620–0.752 | 4.491         | 0.605       | 0.729       |
| <b>the Normal Group vs. the FFKC Group</b> |                                     |       |             |               |             |             |
| Scheimpflug                                | TBI                                 | 0.665 | 0.598–0.733 | 0.265         | 0.561       | 0.679       |
| + Air-puff tonometry                       |                                     |       |             |               |             |             |
| Scheimpflug tomography                     | PRFI                                | 0.670 | 0.599–0.741 | 0.165         | 0.745       | 0.536       |
| Scheimpflug tomography                     | BAD-D                               | 0.661 | 0.592–0.730 | 1.465         | 0.756       | 0.571       |
| Scheimpflug tomography                     | RPIMax                              | 0.661 | 0.592–0.730 | 1.465         | 0.756       | 0.571       |
| SD-OCT                                     | EPI 5 mm SN-IT                      | 0.595 | 0.523–0.668 | 0.405         | 0.790       | 0.405       |
| SD-OCT                                     | EPI 5 mm Superior-Inferior          | 0.593 | 0.520–0.666 | −0.605        | 0.683       | 0.500       |
| SD-OCT                                     | EPI 5 mm S-I                        | 0.575 | 0.502–0.647 | −2.570        | 0.413       | 0.738       |
| Air-puff tonometry                         | A2_Deflection_Amp_mm                | 0.734 | 0.668–0.801 | 0.133         | 0.738       | 0.702       |
| Air-puff tonometry                         | A2_Deflection_Area_mm <sup>-1</sup> | 0.736 | 0.669–0.803 | 0.332         | 0.727       | 0.714       |
| Air-puff tonometry                         | CBI beta                            | 0.692 | 0.626–0.758 | −0.352        | 0.661       | 0.690       |

AUC = area under the receiver operating characteristic curve; AKC = advanced keratoconus; EKC = early keratoconus; SD-OCT = spectral domain optical coherence tomography; TBI = the tomographic and biomechanical index; BADD = the Belin-Ambrósio deviation index; ISV = the index of surface variance; PRFI = the Pentacam random forest index; IHD = the index of height decentration; IVA = the index of vertical asymmetry; RPIMax = the maximum pachymetric progression index; EPI = Epithelium; SN = superior nasal; IT = inferior temporal; S = Superior; I = Inferior; CBI beta = the linear Corvis biomechanical index; CBI = the Corvis biomechanical index.

**Table 3**

Current Established Artificial Intelligence Models' Performance.

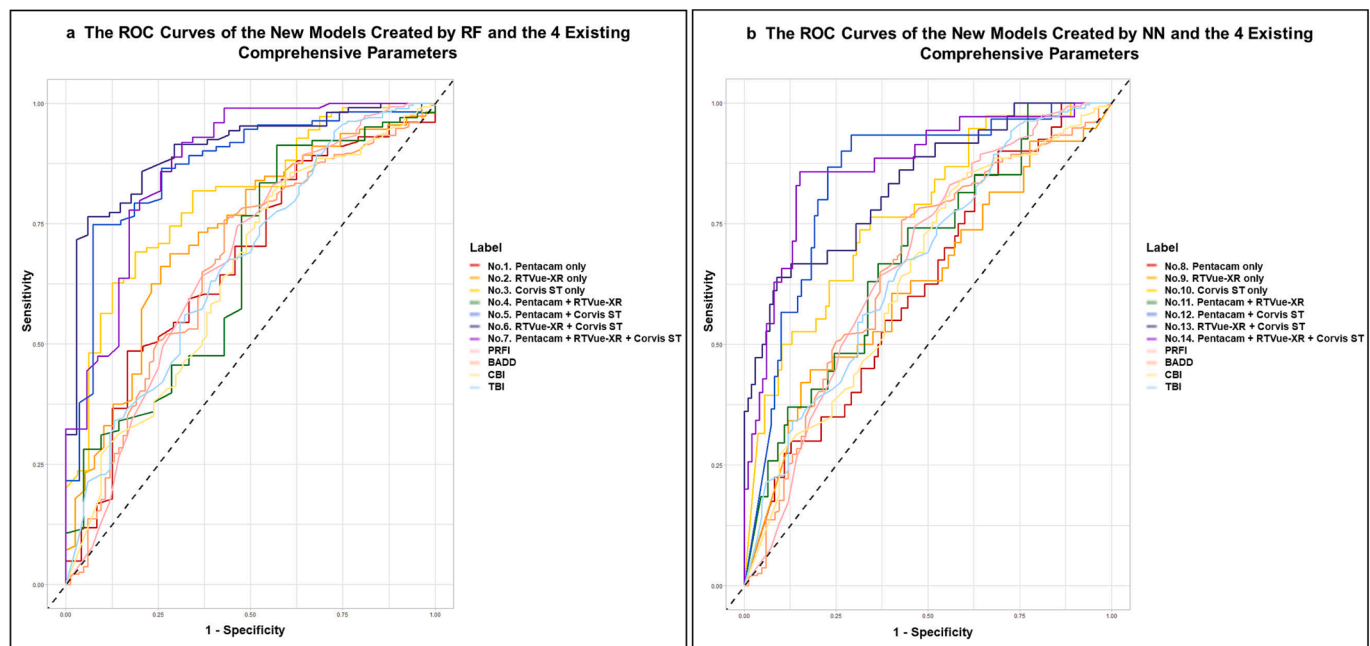
| No.                                             | Included features               | Accuracy | Sensitivity | Specificity | AUC   |
|-------------------------------------------------|---------------------------------|----------|-------------|-------------|-------|
| <b>a: Models established by Random Forest</b>   |                                 |          |             |             |       |
| 6                                               | RTVue-XR + Corvis ST            | 84.29%   | 73.53%      | 87.74%      | 0.902 |
| 7                                               | Pentacam + RTVue-XR + Corvis ST | 84.33%   | 68.57%      | 89.90%      | 0.871 |
| 5                                               | Pentacam + Corvis ST            | 84.06%   | 59.26%      | 90.09%      | 0.867 |
| 3                                               | Corvis ST only                  | 77.46%   | 62.50%      | 81.82%      | 0.801 |
| 2                                               | RTVue-XR only                   | 74.83%   | 28.21%      | 91.07%      | 0.725 |
| 1                                               | Pentacam only                   | 72.00%   | 45.83%      | 78.22%      | 0.666 |
| 4                                               | Pentacam + RTVue-XR             | 81.45%   | 42.86%      | 89.32%      | 0.658 |
| <b>b: Models established by Neutral Network</b> |                                 |          |             |             |       |
| 14                                              | Pentacam + RTVue-XR + Corvis ST | 85.07%   | 71.43%      | 86.87%      | 0.869 |
| 12                                              | Pentacam + Corvis ST            | 82.14%   | 56.67%      | 89.09%      | 0.842 |
| 13                                              | RTVue-XR + Corvis ST            | 84.06%   | 63.89%      | 91.18%      | 0.832 |
| 10                                              | Corvis ST only                  | 77.40%   | 50.00%      | 87.04%      | 0.770 |
| 11                                              | Pentacam + RTVue-XR             | 72.99%   | 40.74%      | 80.90%      | 0.681 |
| 9                                               | RTVue-XR only                   | 72.90%   | 36.84%      | 84.61%      | 0.625 |
| 8                                               | Pentacam only                   | 68.00%   | 30.00%      | 81.82%      | 0.620 |

No. = number; AUC = area under the receiver operating characteristic curve.

combination of air-puff tonometry with Scheimpflug, emphasizing the input of the epithelial and stromal profiles provided by OCT.

Various AI algorithms were built based on either a single device or combinations of devices, each claiming excellent KC diagnostic abilities [32]. The robustness of these approaches was unclear since studies rarely compare all available device types. For clinicians, it is difficult to determine the most useful tool for KC screening. Furthermore, diverse inclusion criteria between studies, the low number of early-stage KC patients included, and the wide range of devices and indices being used, highlight the need for comparative analyses. In this study, a large number of KC patients was included using the same inclusion criteria as the original studies that developed the PRFI, CBI, and TBI parameters. Moreover, three different devices based on different measuring principles were differently combined to conduct a comprehensive and objective analysis.

The current results confirmed that existing parameters (PRFI, BADD, and CBI), based on a single device (Scheimpflug tomography or air-puff tonometry), can accurately diagnose EKC and AKC, but are less reliable for FFKC. Hence, our models focused mainly on distinguishing FFKC from normal eyes. Models 1 and 8 (Pentacam only) performed similarly to PRFI and BADD, suggesting the latter may have reached peak efficiency. Meanwhile, Models 3 and 10 (Corvis only) outperformed CBI, suggesting that CBI still has room for improvement. This may be because CBI was based on logistic regression rather than RF or NN, and the



**Fig. 2.** Comparison of seven artificial intelligence models created by random forest (a) and neural networks (b) and existing comprehensive parameters with receiver operating characteristic (ROC) curve analysis. PRFI = the Pentacam random forest index; BADD = the Belin-Ambrósio deviation index; CBI = the Corvis biomechanical index; TBI = the tomographic and biomechanical index.

features of CBI were selected by forward stepwise inclusion [8], which may have overlooked other potentially useful characteristics.

TBI, which combines Scheimpflug with air-puff tonometry, did not obtain superior FFKC diagnostic results compared to single-device parameters. This suggests that TBI, in its current iteration, is not yet fully optimised for FFKC diagnosis. However, Models 5 and 12 (based on Scheimpflug and air-puff tonometry) outperformed TBI, PRFI, BADD, CBI, as well as the single-device models in the current study (Models 1–3, Models 8–10). This also suggests that TBI may require further iterations to grasp forme fruste biomechanics. Although no detailed formula for TBI were provided in the original study [19], it is known that TBI was built based on RF without feature selection, which might have left its diagnostic ability below potential.

To the best of our knowledge, this was the first attempt to combine three devices with different measuring principles to diagnose FFKC with the help of AI. This three-devices combination obtained the highest accuracy (Model 7), although it was only marginally better than air-puff tonometry with SD-OCT, which may be explained by the aforementioned fact: the combination of two corneal shape measuring devices did not improve FFKC diagnosis probably because they provide much the same information. While any of these devices could accurately diagnose EKC and AKC on their own, the combination of air-puff tonometry with either Scheimpflug tomography or SD-OCT led to an obvious improvement in FFKC diagnostic ability. Combining all three devices however, did not obviously improve the diagnostic ability for FFKC.

From health economics and efficiency perspective [33], clinicians are suggested to choose their KC examination devices based on their target population. For patients with high risk of KC, *e.g.*, age between 12 and 25, with KC family history, eye rubbing, LVC candidates, thin cornea, or corneal astigmatism more than 1.5 D [34,35], a combination of air-puff tonometry with Scheimpflug tomography/ SD-OCT to detect FFKC is recommended. For patients with low KC risk, the diagnostic ability of any single clinical device may be sufficient.

One limitation of the current study may be that it only included a selection of commercially available devices. As such, it is unclear whether combinations with other devices of the same type, *e.g.*, the self-combined Placido/SD-OCT device (MS-39, CSO Italia, Firenze, Italy), perform better. Further investigations need to be conducted in this

regard.

In conclusion, existing AI methods can accurately diagnose EKC and AKC but their diagnostic ability of FFKC is low. Compared to using a single device, combining air-puff tonometry with Scheimpflug tomography or SD-OCT can vastly improve the diagnostic efficiency for FFKC but simultaneous use of three devices offers no additional advantages. Clinicians and optometrists are recommended to choose a combination of corneal biomechanical and morphological biometry, preferably including the epithelial profile, to improve their FFKC diagnosis in patients with high risk of KC.

#### Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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#### Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.clae.2023.101840>.

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